Deploy Application using Ansible + CICD pipeline via Github Actions

Create project folder: ansible-cicd

Move to the folder and create folder ansible

Create 2 files under ansible folder

Inventory.yml

```
all:
  hosts:
    target_host:
      ansible_host: "{{ host_placeholder }}"
      ansible_user: "{{ user_placeholder }}"
      ansible_ssh_private_key_file: ~/.ssh/id_rsa
```

create playbook.yml

```
---
- name: Test Connection with instance and install NgInx
  hosts: all
  become: yes
  tasks:
    - name: Ensure Nginx is installed
      apt:
        name: nginx
        state: present
        update_cache: yes

    - name: Start and Enable Nginx
      service:
        name: nginx
        state: started
        enabled: yes
```

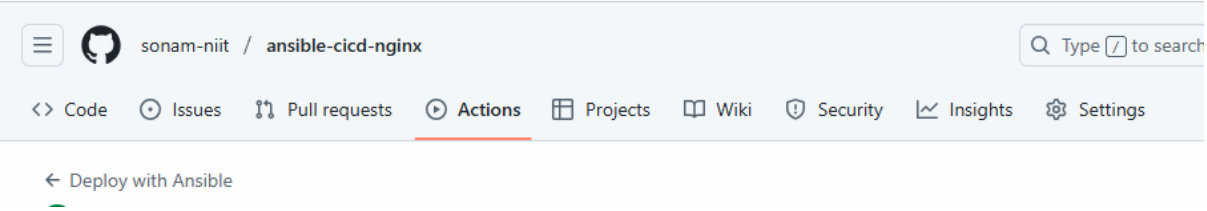
Create .github/workflows/deploy.yml

```
name: Deploy with Ansible
on:
    push:
```

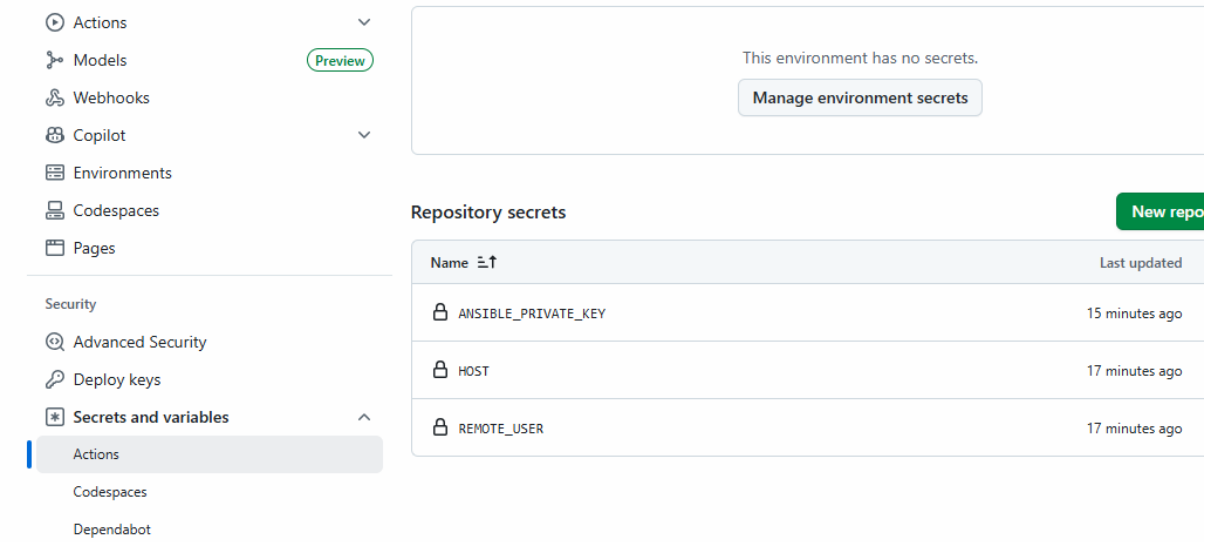
```yaml
        branches:
            - main

jobs:
    deploy:
        runs-on: ubuntu-latest

        steps:
            - name: Checkout Repo
              uses: actions/checkout@v4

            - name: Set up Python and
Ansiblec:\Users\NEW\Downloads\testsonam.pem
              run: |
                sudo apt update
                sudo apt install -y python3-pip
                pip3 install ansible

            - name: Add SSh key and set permission
              run: |
                mkdir -p ~/.ssh
                echo "${{ secrets.ANSIBLE_PRIVATE_KEY }}" >
~/.ssh/id_rsa
                chmod 600 ~/.ssh/id_rsa
                ssh-keyscan -H ${{ secrets.HOST }} >>
~/.ssh/known_hosts

            - name: Replace Placeholders in my inventory
              run: |
                sed -i "s/{{ host_placeholder }}/${{
secrets.HOST }}/" ansible/inventory.yml
                sed -i "s/{{ user_placeholder }}/${{
secrets.REMOTE_USER }}/" ansible/inventory.yml

            - name: Run Ansible Playbook
              run: |
                ansible-playbook ansible/playbook.yml -i
ansible/inventory.yml
```

Now Very important setup your github Repository

Go to your repo settings

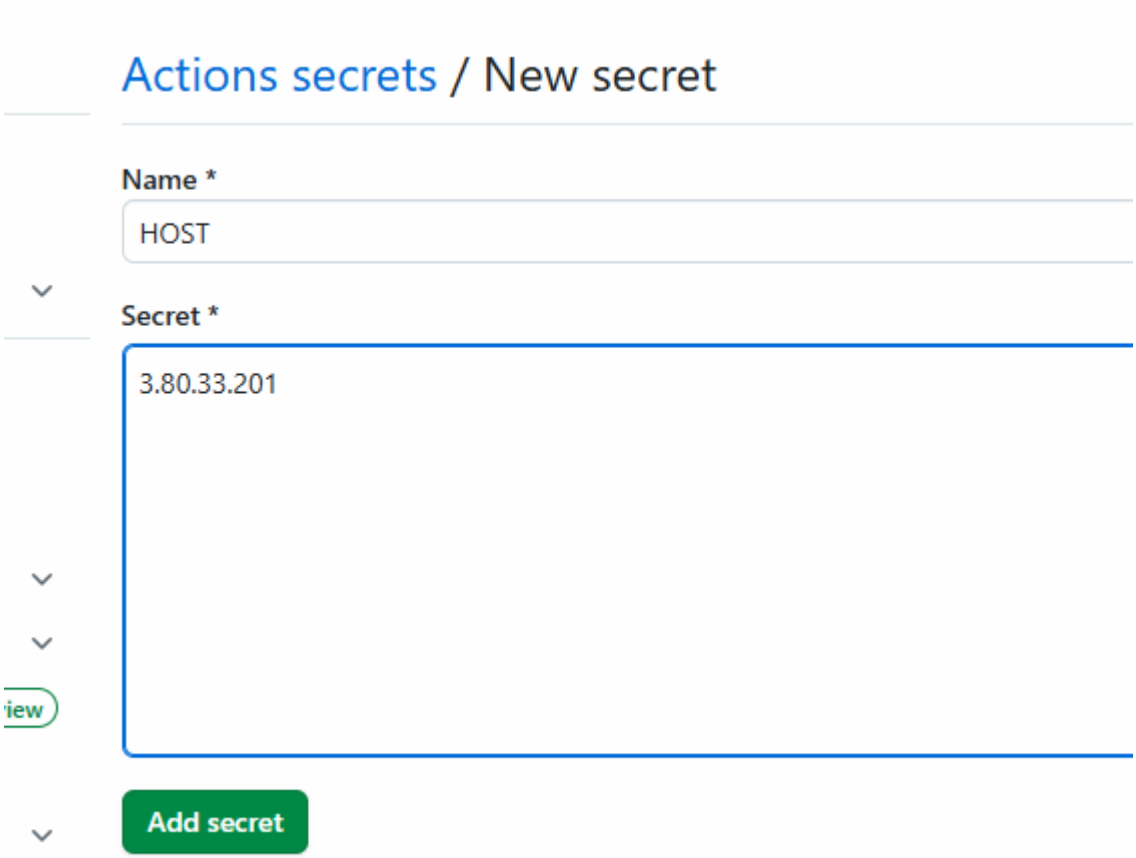


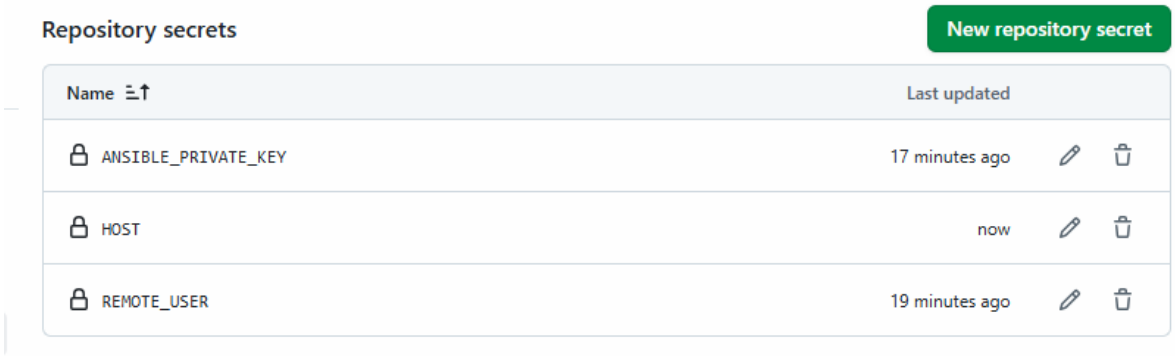
Click on settings



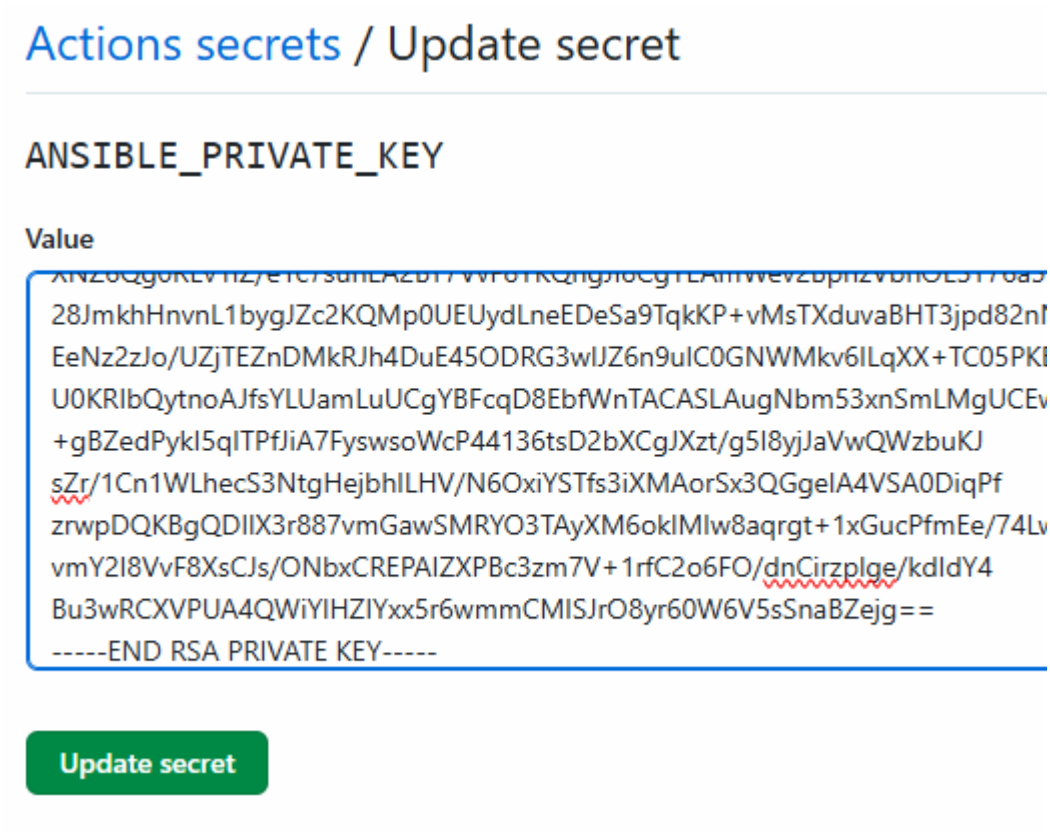
Click on secrets and variables

Actions → create new Repository secret





For Key you can copy paste the content of your .pem file



Then just do git push.

Check Actions

For entire project structure and output you can check below repo link

deploy
succeeded 10 minutes ago in 1m 53s

Search

Run Ansible Playbook

```
4
5  PLAY [Test Connection with instance and install NgInx] *************************
6
7  TASK [Gathering Facts] *********************************************************
8  Warning: : Platform linux on host target_host is using the discovered Python
9  interpreter at /usr/bin/python3.12, but future installation of another Python
10 interpreter could change the meaning of that path. See
11 https://docs.ansible.com/ansible-
12 core/2.18/reference_appendices/interpreter_discovery.html for more information.
13 ok: [target_host]
14
15 TASK [Ensure Nginx is installed] *********************************************
16 changed: [target_host]
17
18 TASK [Start and Enable Nginx] *************************************************
19 ok: [target_host]
20
21 PLAY RECAP *********************************************************************
22 target_host                : ok=3    changed=1    unreachable=0    failed=0    skipped=0
23
```

If all executed successfully then you can see the default page of NgInx with public IP (make sure port 80 is open in your security group)

Reference Repo Link: https://github.com/sonam-niit/Devops-Sept-2025-Ansible-CICD